

Technical Specification: Advanced CGM Membrane Design (Rev. 9)

1. Executive Summary & Wear Classification

This technical specification establishes a covalently tethered bio-interface membrane architecture for implantable Continuous Glucose Monitoring (CGM) sensors (90-day wear). Beyond baseline physical parameters and corrected ISO 10993-18 Annex E E&L evaluations, this revision directly addresses critical in vivo systemic risks: managing secondary mass transport kinetics, ensuring multi-layer interfacial durability against dynamic foreign body responses (FBR), and bridging physical limits with software-driven, glucose-range-dependent recalibration for extreme phenotypic states.

2. Comprehensive Membrane Architecture & Material Specifications

Property / Parameter	Design Specification	Measurement Method / Standard
Total Membrane Thickness	$20 \pm 2 \text{ }\mu\text{m}$	Cross-sectional SEM / Micrometer Profilometry
SBMA Outer Layer Thickness	$1.5 \pm 0.3 \text{ }\mu\text{m}$	X-ray Photoelectron Spectroscopy (XPS) Depth Profiling
Mesh Size / Correlation Length (ξ)	$3.2 \pm 0.4 \text{ nm}$	Fluorescence Recovery After Photobleaching (FRAP)
Equilibrium Swelling Ratio (SR)	$165 \pm 8 \text{ }\%$	Gravimetric Swelling in PBS at 37°C
Equilibrium Water Content (EWC)	$70 \pm 3 \text{ }\%$	Desiccation Weight Loss Analysis
Young's Modulus (E)	$1.1 \pm 0.15 \text{ MPa}$	Dynamic Mechanical Analysis (DMA) / Tensile Testing
Tensile Strength & Elongation	1.3 MPa / 175% break strain	ASTM D638 Micro-Tensile Testing
Oxygen Diffusion Coeff. (D_{O_2})	$2.1 \times 10^{-5} \text{ cm}^2/\text{s}$	Electrochemical Permeation Cell (37°C, PBS)
Glucose Diffusion Coeff. (D_{glu})	$1.8 \times 10^{-6} \text{ cm}^2/\text{s}$	Diffusion Chamber / HPLC Assay (37°C, PBS)
Enzyme Loading & Interface	150 U/cm ² (GOx)	Covalently immobilized at inner working electrode interface

3. Systems-Level Permeability & Transport Validation

The core functional requirement to prevent macroscopic oxygen limitation ("oxygen deficit") is established via the steady-state permeability ratio:

$$P_{\text{O}_2} / P_{\text{glucose}} = 116.7 : 1$$

In Vivo Kinetic Caveats: While the static 116.7:1 ratio acts as a robust primary design indicator, absolute sensor accuracy is modulated by microenvironmental interstitial $p\text{O}_2$ variability, enzyme apparent Michaelis-Menten kinetics (K_M^{app}), and secondary hydrogen peroxide (H_2O_2) clearance rates to the working electrode.

4. Interfacial Durability & Bio-Interface Stability (90-Day Wear)

The translation of scientifically plausible antifouling into a 90-day functional reality requires surviving aggressive environments:

- Interfacial Adhesion:** The $1.5 \pm 0.3 \text{ }\mu\text{m}$ SBMA layer is covalently anchored via radical polymerization to underlying PVA-GMA vinyl handles to prevent delamination.
- Sterilization Survival:** Crosslinking density is over-indexed during synthesis to account for Ethylene Oxide (EtO) chain scission, preserving the 3.2 nm mesh size.
- Foreign Body Response (FBR):** The ultrathin dimension prevents the SBMA brush from becoming a parasitic bulk diffusion barrier as tissue remodels.

5. Regulatory E&L Compliance

Regulatory E&L (ISO 10993-18 Annex E): Using the corrected formula for a 90-day implant ($\text{DBT} = 1.5 \text{ g/day}$, $A = 1$, $B = 50 \text{ mL}$, $C = 2$, $UF = 2$):

$$\text{AET} = \frac{1.5 \times 1}{50 \times 2 \times 2} = 0.0075 \text{ g/mL}$$

6. Algorithmic Fusion: Glucose-Range-Dependent Recalibration

To bridge the physical limitations of the membrane during periods of chronic hyperglycemia ($>250 \text{ mg/dL}$), the sensor architecture is paired with a state-dependent adaptive continuous calibration algorithm. Sustained hyperglycemia maximizes GOx turnover, accelerating local H_2O_2 oxidative stress, enzyme passivation, and localized pO_2 depletion, driving the system toward its K_M^{app} boundary.

- Phenotype-Weighted Calibration:** If the patient's continuous glycemic profile (AUC) exceeds threshold limits within the first 72 hours, the system defaults to a distinct calibration profile optimized for high-range linearity to counter high-end signal flattening.
- Dynamic Sensitivity Decay Modeling:** The baseline sensitivity is algorithmically degraded proportional to extreme glucose exposure over time:

$$S_{\text{adj}}(t) = S_{\text{base}} - \alpha \int \max(0, G(\tau) - G_{\text{threshold}}) d\tau$$

Where α is an empirically derived degradation coefficient and $G_{\text{threshold}}$ represents the hyperglycemic boundary.

- Piecewise Error Correction:** Sub-routines automatically adjust gain and offset specifically for current (nA) readings falling within the upper physiological brackets, correcting for expected mixed-order kinetic plateauing before it manifests as clinical error.